

Hierarchical

DANA 4840

In this article we start by describing the agglomerative clustering algorithms. Next, we provide R lab sections with many examples for computing and visualizing hierarchical clustering. We continue by explaining how to interpret dendrogram. Finally, we provide R codes for cutting dendrograms into groups.

Note that, it's generally recommended to standardize variables in the data set before performing subsequent analysis. Standardization makes variables comparable, when they are measured in different scales. For example one variable can measure the height in meter and another variable can measure the weight in kg. The R function `scale()` can be used for standardization, See `?scale` documentation.

```
# Load the data
data("USArrests")

# Standardize the data
df <- scale(USArrests)

# Show the first 6 rows
head(df, nrow = 6)

##           Murder  Assault  UrbanPop      Rape
## Alabama    1.24256408 0.7828393 -0.5209066 -0.003416473
## Alaska     0.50786248 1.1068225 -1.2117642  2.484202941
## Arizona    0.07163341 1.4788032  0.9989801  1.042878388
## Arkansas   0.23234938 0.2308680 -1.0735927 -0.184916602
## California 0.27826823 1.2628144  1.7589234  2.067820292
## Colorado   0.02571456 0.3988593  0.8608085  1.864967207

# Compute the dissimilarity matrix
# df = the standardized data
res.dist <- dist(df, method = "euclidean")
```

Note that, the function `dist()` computes the distance between the rows of a data matrix using the specified distance measure method.

```
as.matrix(res.dist)[1:6, 1:6]

##           Alabama  Alaska  Arizona  Arkansas  California  Colorado
## Alabama    0.000000 2.703754 2.293520 1.289810   3.263110 2.651067
## Alaska     2.703754 0.000000 2.700643 2.826039   3.012541 2.326519
## Arizona    2.293520 2.700643 0.000000 2.717758   1.310484 1.365031
## Arkansas   1.289810 2.826039 2.717758 0.000000   3.763641 2.831051
## California 3.263110 3.012541 1.310484 3.763641   0.000000 1.287619
## Colorado   2.651067 2.326519 1.365031 2.831051   1.287619 0.000000

res.hc <- hclust(d = res.dist, method = "ward.D2")
```

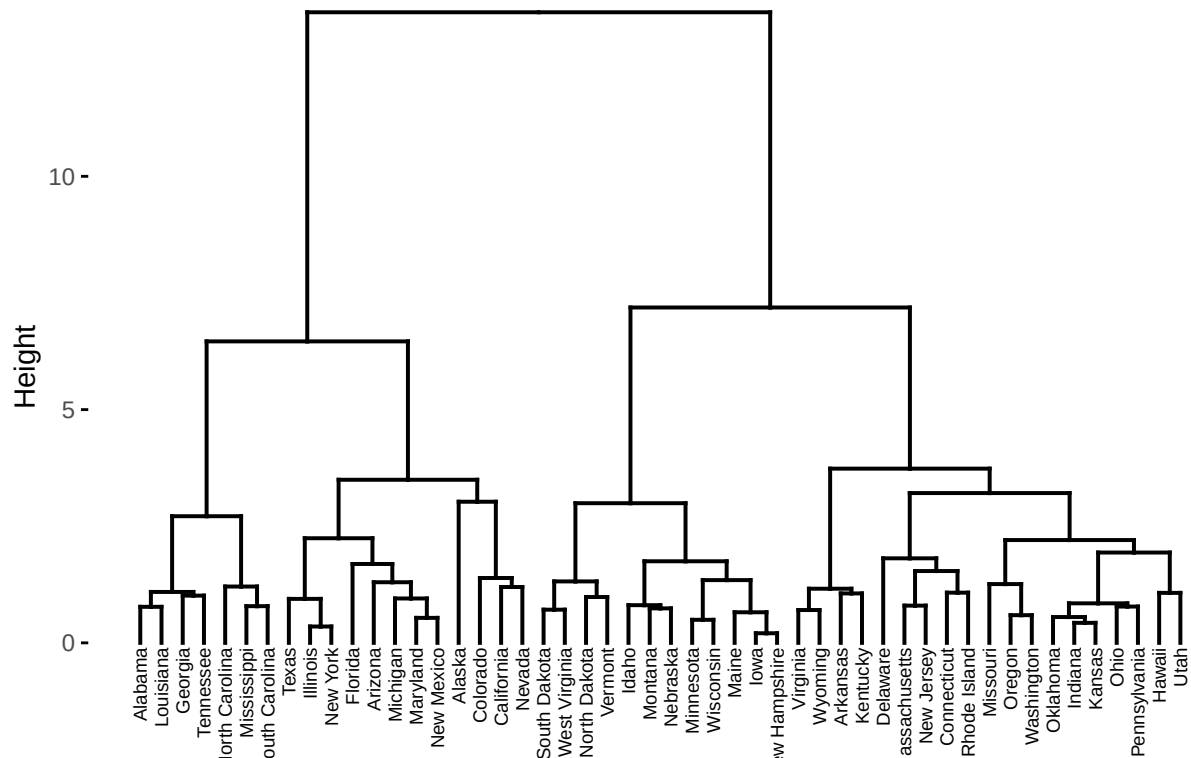
- Maximum or *complete linkage*: The distance between two clusters is defined as the maximum value of all pairwise distances between the elements in cluster 1 and the elements in cluster 2. It tends to produce more compact clusters.

```
## Complete linkage and Ward's method are generally preferred.

## ----visualize-dendrogram, fig.height=4-----
# cex: label size
library("factoextra")

## Warning: package 'factoextra' was built under R version 4.0.5
## Loading required package: ggplot2
## Warning: package 'ggplot2' was built under R version 4.0.5
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
fviz_dend(res.hc, cex = 0.5)
```

Cluster Dendrogram



Note that, conclusions about the proximity of two objects can be drawn only based on the height where branches containing those two objects first are fused. We cannot use the proximity of two objects along the horizontal axis as a criteria of their similarity.

```
# Compute cophentic distance
res.coph <- cophenetic(res.hc)

# Correlation between cophenetic distance and
# the original distance
cor(res.dist, res.coph)

## [1] 0.6975266
```

```

res.hc2 <- hclust(res.dist, method = "average")

cor(res.dist, cophenetic(res.hc2))

## [1] 0.7180382
# Cut tree into 4 groups
grp <- cutree(res.hc, k = 4)
head(grp, n = 4)

## Alabama Alaska Arizona Arkansas
##      1      2      2      3
# Number of members in each cluster
table(grp)

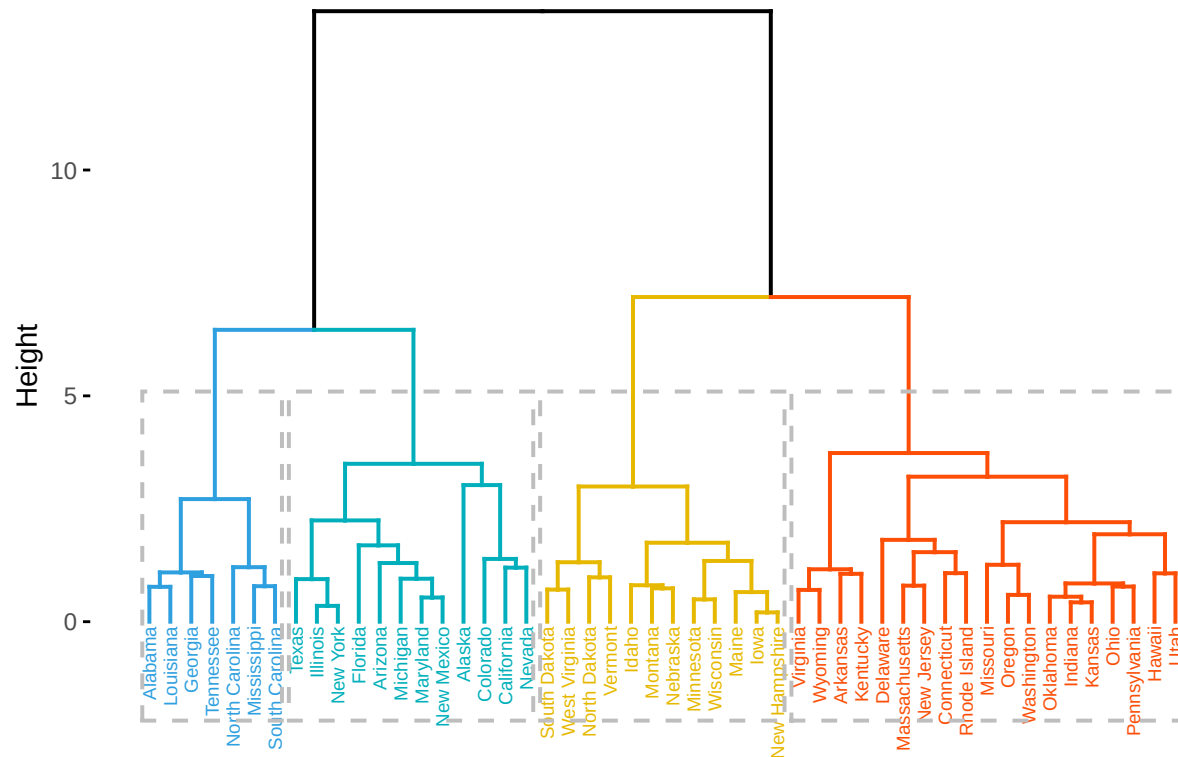
## grp
##  1  2  3  4
##  7 12 19 12
# Get the names for the members of cluster 1
rownames(df)[grp == 1]

## [1] "Alabama"      "Georgia"      "Louisiana"    "Mississippi"
## [5] "North Carolina" "South Carolina" "Tennessee"

## ----cutree-cut-dendrogram, height = 4-----
# Cut in 4 groups and color by groups
fviz_dend(res.hc, k = 4, # Cut in four groups
  cex = 0.5, # label size
  k_colors = c("#2E9FDF", "#00AFBB", "#E7B800", "#FC4E07"),
  color_labels_by_k = TRUE, # color labels by groups
  rect = TRUE # Add rectangle around groups
)

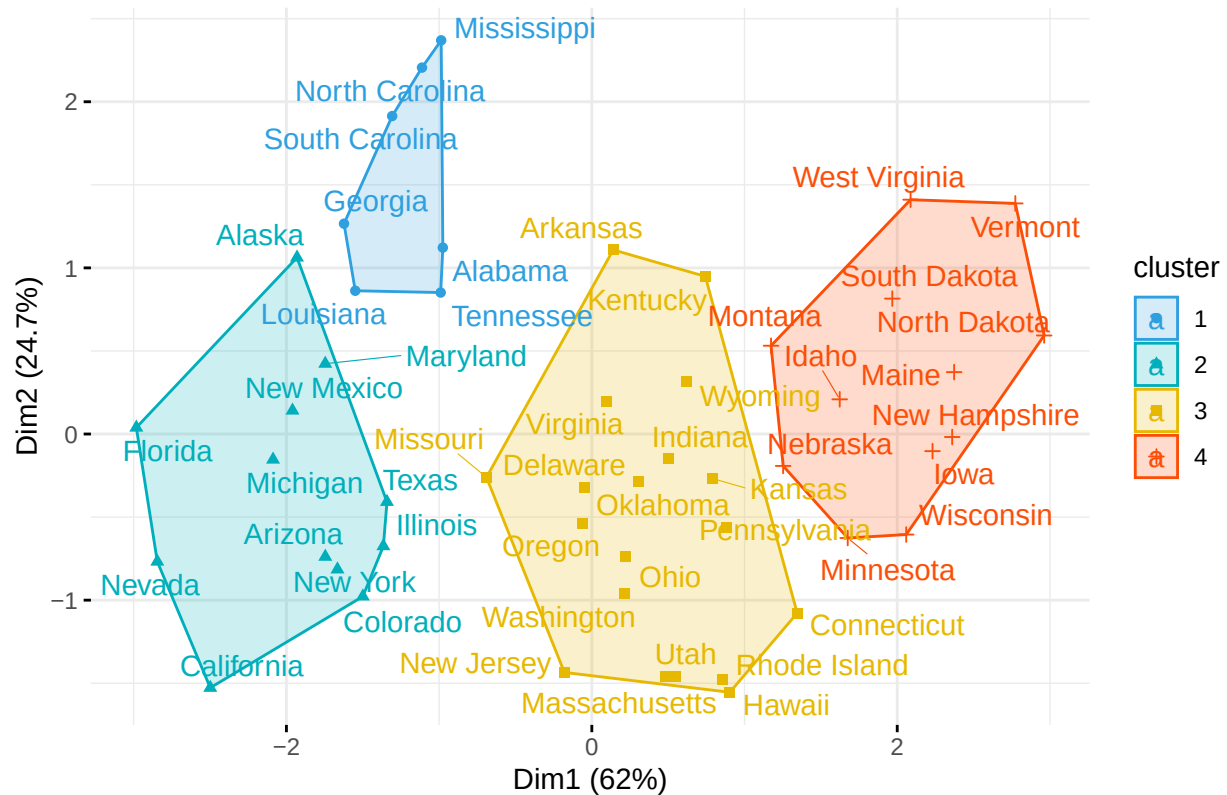
```

Cluster Dendrogram



```
## ----cluster-plot, fig.width=6, fig.height=6-----
fviz_cluster(list(data = df, cluster = grp),
  palette = c("#2E9FDF", "#00AFBB", "#E7B800", "#FC4E07"),
  ellipse.type = "convex", # Concentration ellipse
  repel = TRUE, # Avoid label overplotting (slow)
  show.clust.cent = FALSE, ggtheme = theme_minimal())
```

Cluster plot



```
library("cluster")
## # Agglomerative Nesting (Hierarchical Clustering)

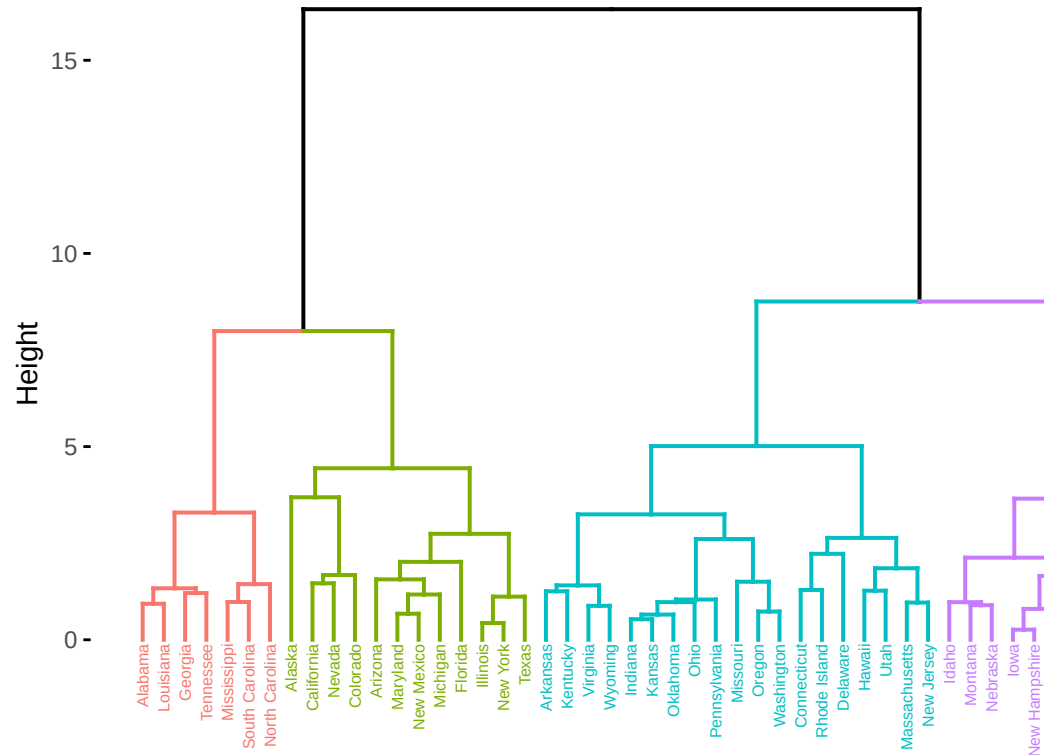
res.agnes <- agnes(x = USArrests, # data matrix
  stand = TRUE, # Standardize the data
  metric = "euclidean", # metric for distance matrix
  method = "ward" # Linkage method
)
```

Agnes hierarchical clustering

```
res.diana <- diana(x = USArrests, # data matrix
  stand = TRUE, # standardize the data
  metric = "euclidean" # metric for distance matrix
)
```

```
fviz_dend(res.agnes, cex = 0.4, k = 4)
```

Cluster Dendrogram

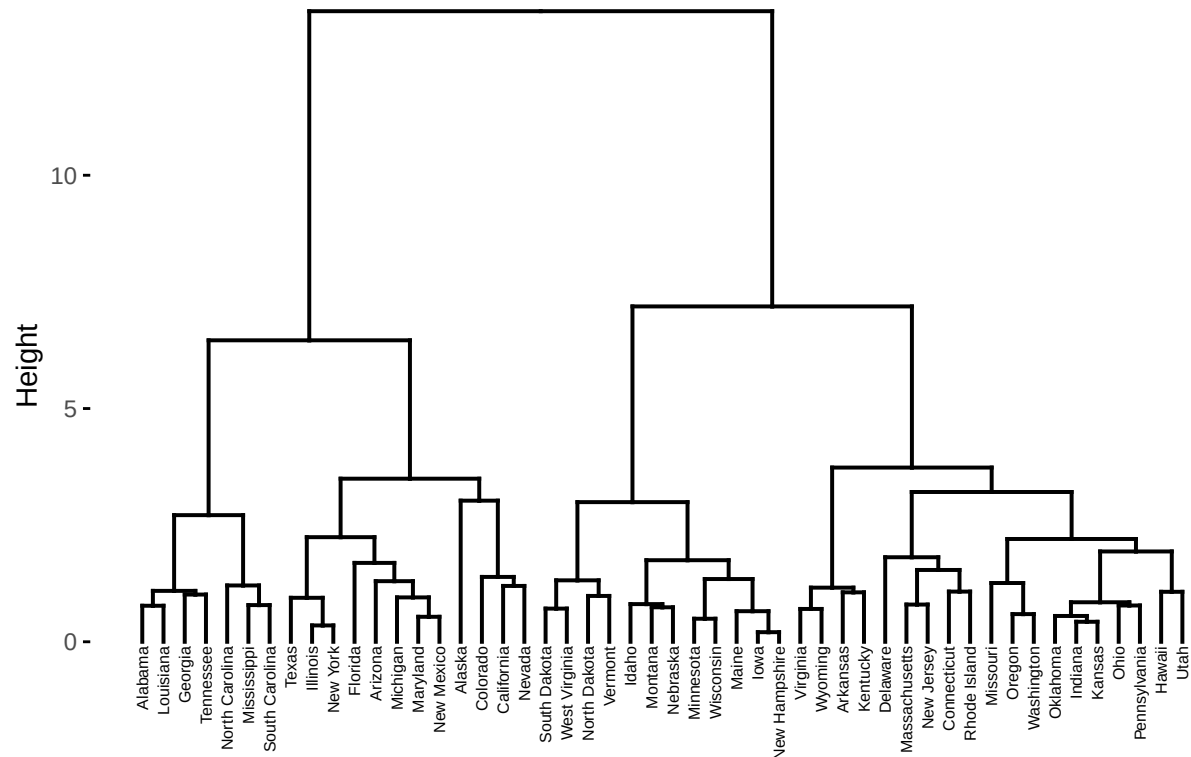


Divisive Analysis Clustering

Note that, when the data are scaled, the Euclidean distance of the z-scores is the same as correlation distance.

```
factoextra::fviz_dend(hclust(dist(scale(USArrests))), "ward.D2"),
  main = "Hierarchical Clustering", sub = "", cex = 0.4)
```

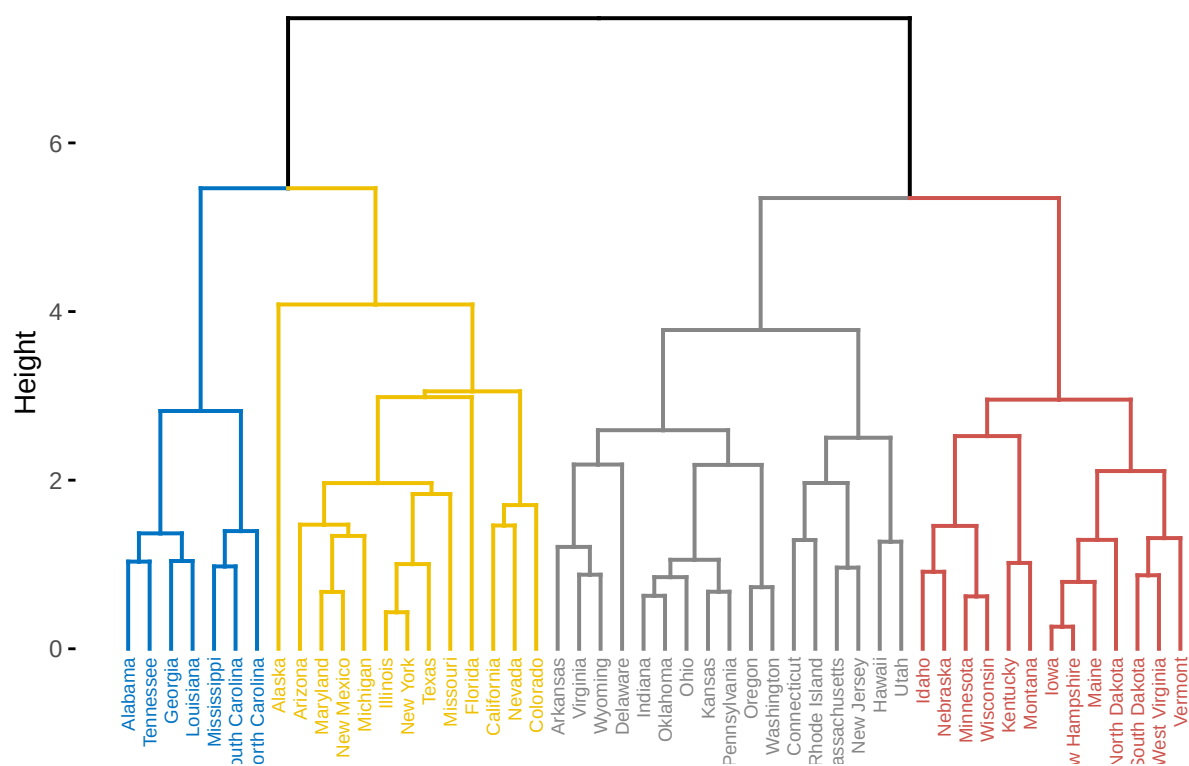
Hierarchical Clustering



```
# Compute diana()
library(cluster)
res.diana <- diana(USArrests, stand = TRUE)

# Plot the dendrogram
library(factoextra)
fviz_dend(res.diana, cex = 0.5,
           k = 4, # Cut in four groups
           palette = "jco" # Color palette
)
```

Cluster Dendrogram



compute diana